Cirrus SR22 G3 / G5 (SR22)

Continental IO-550-N, 310 hp · 4 (G3) / 5 (G5) seats · Typical normally-aspirated SR22 G3 (3,400 lb) and G5 (3,600 lb) POH procedures with Perspective avionics; where the two differ both figures are given (G3 / G5).

V-speeds & limits

Vr	**70 KIAS (G3) / 73 KIAS (G5), flaps 50%**
Vx	**78 KIAS (G3) / 81 KIAS (G5)**
Vy	**101 KIAS (G3) / 108 KIAS (G5)**
Best glide	**88 KIAS (G3) / 92 KIAS (G5)**
Vfe	**50%: 119 KIAS (G3) / 150 KIAS (G5); 100%: 104 KIAS (G3) / 110 KIAS (G5)**
Va	**133 KIAS at 3,400 lb (G3) / 140 KIAS at 3,600 lb (G5)**
Vno	**178 KIAS (G3) / 177 KIAS (G5)**
Vne	**204 KIAS (G3) / 201 KIAS (G5)**
Vpd (max CAPS deploy, demonstrated)	**133 KIAS (G3) / 140 KIAS (G5)**
Vs / Vso	**approx 73 / 60 KIAS (max weight)**
Vref	**flaps 100%: 77–80 KIAS (G3) / 80–85 KIAS (G5); flaps 50%: 85–90 KIAS**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
CAPS handle safety pin	**REMOVED (handle cover in place, pin stowed)**
Passenger briefing	**COMPLETE (belts / airbags, doors, egress, CAPS – who pulls and when)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Doors	**CLOSED & LATCHED**
Parking brake	**SET**
Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Avionics power	**OFF**
Bat 2 master	**ON – voltage CHECK (approx 24 V on essential bus)**
Bat 1 master	**ON**
Alt 1 / Alt 2 masters	**OFF until after start**
Annunciators / CAS messages	**CHECK**
Fuel quantity	**CHECK (gauge and visual)**
Flaps	**UP (check indicator)**
Strobe lights	**ON**

Engine Start

Brakes	**HOLD**
Propeller area	**CLEAR**
Power lever	**FULL OPEN**
Mixture	**FULL RICH**
Fuel pump	**PRIME until fuel flow peaks (approx 2–3 sec), then BOOST**
Power lever	**1/4 OPEN (approx 1/4 in)**

▶ Engine hot or flooded?

No:

Yes:

Fuel pump	**BOOST (do not prime)**
Mixture	**IDLE CUTOFF if flooded – advance to FULL RICH as engine fires**
Power lever	**1/2 OPEN if flooded, then idle as engine fires**
Ignition switch	**START (release when engine fires)**
Power lever	**1000 RPM**
Oil pressure	**CHECK rising within 30 sec (30–60 psi green; 10 psi min idle)**
Alt 1 master	**ON**
Alt 2 master	**ON**
Avionics power	**ON**
Alternator output / voltage	**CHECK both alternators charging (approx 28 V)**
Engine instruments	**CHECK**
Fuel pump	**BOOST (leave on for taxi and takeoff)**

Before Taxi

Flaps	**UP**
PFD / MFD	**SET (baro, nav, flight plan) – compare altimeters**
Fuel pump	**BOOST**
Brakes	**CHECK on initial roll**
Flight instruments	**CHECK during taxi (HSI turns correct, slip-skid, standby instruments agree)**
Autopilot / electric trim	**CHECK & disconnect**
Cabin heat / defrost	**AS REQUIRED**

Before Takeoff

Doors	**LATCHED**
Seats / belts / harnesses	**SECURE**
Fuel selector	**FULLEST TANK**
Fuel pump	**BOOST**
Flaps	**50%**
Trim	**TAKEOFF (elevator and rudder trim indicators in takeoff range)**
Flight controls	**FREE & CORRECT**
Mixture	**FULL RICH (lean per fuel-flow placard above approx 5,000 ft elevation)**
Power lever	**1700 RPM**
Alt 1 / Alt 2	**CHECK charging (ammeters positive, voltage approx 28 V)**
Pitot heat	**CHECK (current rise), then as required**
Magnetos	**CHECK L & R (150 RPM max drop, 75 RPM max difference)**
Engine instruments	**CHECK (oil pressure 30–60 psi, oil temp 100–240°F, CHT in green)**
Power lever	**IDLE check, then 1000 RPM**
Avionics / autopilot	**SET – autopilot OFF, flight director as required**
Transponder	**ALT**
Lights	**AS REQUIRED (landing light ON)**
Takeoff briefing	**COMPLETE (abort point; engine failure below approx 500 ft AGL – land ahead; above – CAPS available)**

After Takeoff / Climb

Flaps	**UP (above 80–85 KIAS, positive rate)**
Climb power	**FULL THROTTLE / 2700 RPM**
Mixture	**LEAN per fuel-flow placard with altitude**
Fuel pump	**BOOST**
Airspeed	**120 KIAS (cruise climb) / Vy 101 (G3) 108 (G5) for best rate**
Engine instruments	**CHECK (CHT below 420°F)**
Landing light	**OFF above pattern altitude (as required)**

Cruise

Cruise power	**SET (55–85% per POH cruise table)**
Mixture	**LEAN per POH cruise table / lean assist**
Fuel pump	**OFF (BOOST when switching tanks or if fuel flow fluctuates)**
Fuel selector	**SWITCH TANKS to keep within 10 gal imbalance – note time**
Fuel quantity / totaliser	**CHECK against plan**
Engine instruments	**CHECK (oil temp 240°F max, CHT in green)**
Electrical	**CHECK (Alt 1 / Alt 2 ammeters, essential bus voltage)**
Altimeter / heading	**CHECK**

Descent

Altimeter	**SET**
Fuel selector	**FULLEST TANK**
Fuel pump	**BOOST**
Mixture	**ENRICH as required**
Power	**REDUCE gradually (keep CHT in green)**
Approach briefing	**COMPLETE**
Seat belts / harnesses	**SECURE**

Before Landing

Fuel selector	**FULLEST TANK**
Fuel pump	**BOOST**
Mixture	**FULL RICH (or as required for field elevation)**
Flaps	**50% below 119 (G3) / 150 (G5) KIAS; 100% below 104 (G3) / 110 (G5) KIAS**
Autopilot	**OFF (no later than minimums / short final)**
Airspeed	**flaps 100%: 77–80 KIAS (G3) / 80–85 KIAS (G5); flaps 50%: 85–90 KIAS**
Landing light	**ON**
Transponder	**ALT**

After Landing

Flaps	**UP**
Fuel pump	**OFF**
Pitot heat	**OFF**
Landing light / strobes	**AS REQUIRED**
Transponder	**STBY / as required**

Shutdown

Parking brake	**SET**
Fuel pump	**OFF**
Avionics power	**OFF**
Power lever	**IDLE**
Mixture	**IDLE CUTOFF**
Ignition switch	**OFF (after propeller stops)**
Lights	**OFF**
Alt 1 / Alt 2 / Bat 1 / Bat 2 masters	**OFF**
CAPS handle safety pin	**INSTALLED**
Chocks / tiedowns / pitot cover	**AS REQUIRED**

Abnormal procedures

Alternator Failure (ALT 1 / ALT 2 caution)

Affected alternator master ..... **OFF, then ON (reset once)**

Ammeter / voltage ..... **CHECK**

▶ Which alternator failed?

Alt 1 (main bus):

- Alt 1 master ..... **OFF**
- Non-essential load ..... **REDUCE – Alt 2 alone is small (approx 20 A on G3; larger on G5)**
- Essential bus voltage ..... **MONITOR**
- Deferred: Alt 1 failed – reduced electrical load on Alt 2, land as soon as practical. ..... **NOTED**

Alt 2 (essential bus):

- Alt 2 master ..... **OFF**
- Essential bus ..... **CONFIRM fed from main bus / Alt 1**
- Deferred: Alt 2 failed – essential bus on Alt 1 only, land as soon as practical. ..... **NOTED**

Both:

- Alt 1 / Alt 2 masters ..... **OFF**
- Non-essential load ..... **SHED (landing light, air conditioning, non-essential avionics)**
- Battery endurance ..... **PLAN approx 30 min at reduced load**
- Deferred: Both alternators failed – battery only, land as soon as possible, flaps before battery depleted. ..... **NOTED**

Low Oil Pressure

OIL PRESS warning or indication below 30 psi

Oil pressure / oil temperature ..... **CHECK and cross-check**

▶ Oil temperature rising or pressure below 10 psi?

Yes – genuine:

- Power ..... **REDUCE to minimum required**
- Airport ..... **DIVERT to nearest suitable – stay within glide of a landing area**
- CAPS ..... **PLAN deployment if engine fails over unsuitable terrain**
- Deferred: Low oil pressure – land as soon as possible, expect possible engine failure. ..... **NOTED**

No – gauge suspect:

- Flight ..... **CONTINUE, monitor oil temperature and pressure**
- Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

High Oil / Cylinder Head Temperature

Mixture ..... **ENRICH**

Airspeed ..... **INCREASE (lower pitch attitude)**

Power ..... **REDUCE**

▶ Temperature returning to green?

Yes:

- Flight ..... **CONTINUE, monitor**

No:

- Oil pressure ..... **MONITOR for drop**
- Deferred: High engine temperature – land as soon as practical, cooling / baffling check required. ..... **NOTED**

Rough Running / Fuel Flow Fluctuation

Fuel pump ..... **BOOST**

Fuel selector ..... **SWITCH TANKS**

Mixture ..... **ADJUST for smooth running**

Alternate induction air ..... **ON**

Magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (CHT, EGT, fuel flow)**

▶ Engine running smoothly?

Yes:

Deferred: Rough running cleared – leave fuel pump BOOST; report to maintenance. ..... **NOTED**

No:

Power ..... **REDUCE to smoothest setting**

CAPS ..... **PLAN deployment if engine fails over unsuitable terrain**

Deferred: Engine rough – land as soon as practical at nearest suitable airport. ..... **NOTED**

Flap Failure

Flap switch ..... **CYCLE once to the desired position**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – flaps up approach 90–95 KIAS, longer runway**

Deferred: Flaps inoperative – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

Door Open in Flight

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 100–110 KIAS**

Belts / loose items ..... **SECURE**

Do not attempt to close the door while flying the airplane. Expect increased noise; handling is essentially normal.

Deferred: Door unlatched – return and land as soon as practical, normal approach speed. ..... **NOTED**

PFD / AHRS / Air Data Failure

Red X on attitude, airspeed, altitude or heading

Standby instruments ..... **CROSS-CHECK and FLY (standby attitude, airspeed, altimeter, compass)**

Display reversion ..... **SELECT (MFD shows PFD data) if PFD screen failed**

AHRS / ADC source ..... **SWAP to the second unit if fitted (Perspective sensor key)**

Autopilot ..... **OFF if misbehaving (may disconnect automatically)**

Pitot heat ..... **ON (if airspeed suspect)**

IMC ..... **EXIT / AVOID**

Deferred: Flight display / sensor failure – partial panel; prefer VMC approach. ..... **NOTED**

Inadvertent Icing

Pitot heat ..... **ON**

Ice protection (TKS) ..... **MAX / HIGH if fitted**

Exit icing conditions ..... **TURN / CLIMB / DESCEND**

Alternate induction air ..... **ON (if induction icing suspected)**

Power ..... **INCREASE to maintain airspeed**

Autopilot ..... **OFF (monitor control forces)**

Landing ..... **FLAPS 50% maximum, approach speed +10–15 KIAS, longer runway**

Deferred: Residual airframe ice – flaps 50% landing, increased approach speed and landing distance. ..... **NOTED**

Static Source Blocked

Altimeter / VSI frozen or airspeed erratic

Pitot heat ..... **ON**

Alternate static source ..... **ON (valve under instrument panel)**

With alternate static selected, expect small airspeed and altimeter errors; cabin vents and heater affect the readings.

Standby instruments / GPS altitude ..... **CROSS-CHECK**

Deferred: Alternate static in use – allow for airspeed and altimeter error on approach. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Power lever (memory) ..... **IDLE**

Brakes (memory) ..... **AS REQUIRED – stop straight ahead**

Mixture ..... **IDLE CUTOFF**

Fuel selector ..... **OFF**

Ignition switch ..... **OFF**

Bat 1 / Bat 2 masters ..... **OFF**

Engine Failure Immediately After Takeoff

Airborne at low altitude

LAND IMMEDIATELY

Airspeed (memory) ..... **88 KIAS (G3) / 92 KIAS (G5) best glide**

▶ Altitude above approx 500 ft AGL?

No – land straight ahead:

Landing site ..... **STRAIGHT AHEAD with small heading changes only**

Flaps ..... **100% when landing assured**

Mixture ..... **IDLE CUTOFF**

Fuel selector ..... **OFF**

Ignition switch ..... **OFF**

Bat 1 / Bat 2 masters ..... **OFF (after flaps set)**

Do not attempt a turn back to the runway.

Yes – CAPS available:

If a safe landing straight ahead is not assured, deploy CAPS now – do not wait for a better position. See CAPS Deployment.

CAPS handle ..... **PULL (if landing ahead not assured)**

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	**88 KIAS (G3) / 92 KIAS (G5) best glide**
Fuel selector (memory)	**SWITCH TANKS**
Fuel pump (memory)	**BOOST**
Mixture (memory)	**FULL RICH, then lean as required**
Alternate induction air (memory)	**ON**
Ignition switch (memory)	**BOTH – check**

▶ Restart attempt practicable?

Yes – restart:

Power lever	**1/2 OPEN**
Ignition switch	**START if propeller has stopped**

▶ Engine restarted?

Yes:

Power	**AS REQUIRED**
Fuel pump	**BOOST**
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system.	**NOTED**

No:

Continue below: forced landing or CAPS.

No:

▶ Safe forced-landing site assured within glide?

Yes – forced landing:

Seats / belts / harnesses	**SECURE**
Flaps	**100% when landing assured**
Airspeed	**80–85 KIAS on final**
Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Ignition switch	**OFF**
Bat 1 / Bat 2 masters	**OFF (after flaps set)**
Touchdown	**MINIMUM SPEED, wings level**

No – deploy CAPS:

CAPS is the primary response to engine failure over terrain where a safe forced landing is not assured. Deploy above the 2,000 ft AGL decision altitude where possible – see CAPS Deployment.

CAPS handle	**PULL**

CAPS Deployment

Loss of control, mid-air collision, structural failure, pilot incapacitation, or engine failure over terrain where a safe forced landing is not assured

- Airspeed (memory) — **MINIMISE (below Vpd 133 KIAS G3 / 140 KIAS G5 if possible)**
- CAPS handle (memory) — **PULL – remove cover, pull straight down with both hands, full force**
- Mixture (memory) — **IDLE CUTOFF**
- Fuel selector (memory) — **OFF**
- Bat 1 / Bat 2 / Alt 1 / Alt 2 masters (memory) — **OFF**
- Seat belts / harnesses (memory) — **TIGHT**
- Brace (memory) — **EMERGENCY LANDING BODY POSITION (hands on legs or arms crossed, chin down)**

CAPS is the primary response to loss of control, mid-air collision, structural failure, pilot incapacitation and engine failure over terrain where a safe forced landing is not assured. Deploy without hesitation in these cases.

Altitude: minimum demonstrated deployment 400 ft AGL in level flight; recovery from loss of control demonstrated at 920 ft AGL. Recommended decision altitude 2,000 ft AGL. Below these altitudes deployment may still be the best option – pull anyway.

Do not deploy for engine failure at altitude when a safe forced landing site is assured – but do not delay below 2,000 ft AGL if it is not. Expect a touchdown about 1,700 ft/min descent; some structural damage is normal.

- After touchdown – doors — **UNLATCH / OPEN**
- Aircraft — **EVACUATE once stopped (beware of canopy dragging in wind)**
- ELT — **ON (if not automatic)**

Inadvertent Spin / Loss of Control

Spin entry or departure from controlled flight

- CAPS handle (memory) — **PULL**

CAPS is the only approved and demonstrated spin recovery method for the SR22. Do not delay to attempt conventional recovery – altitude is being lost at approx 900 ft/turn.

- Power lever — **IDLE**
- Flight controls — **NEUTRAL**
- Complete — **CAPS Deployment procedure**

Engine Fire in Flight

LAND IMMEDIATELY

- Mixture (memory) — **IDLE CUTOFF**
- Fuel selector (memory) — **OFF**
- Fuel pump (memory) — **OFF**
- Power lever (memory) — **IDLE**
- Cabin heat / defrost — **OFF**
- Airspeed — **INCREASE (approx 100 KIAS or more) to help extinguish fire**
- Ignition switch — **OFF**

▶ Fire out?

Yes:

- Forced landing — **EXECUTE – do not attempt restart (Engine Failure in Flight, forced landing section)**

No:

- CAPS handle — **PULL if fire persists and altitude permits (CAPS Deployment)**
- Bat 1 / Bat 2 masters — **OFF after flaps set**

Engine Fire During Start

Mixture (memory)	**IDLE CUTOFF**
Fuel pump (memory)	**OFF**
Fuel selector (memory)	**OFF**
Power lever (memory)	**FULL OPEN**
Ignition switch (memory)	**CONTINUE CRANKING to draw fire into engine**

▶ Fire out?

Yes:

Ignition switch	**OFF**
Engine	**INSPECT before any further start attempt**

No:

Ignition switch	**OFF**
Bat 1 / Bat 2 masters	**OFF**
Fire extinguisher	**OBTAIN / USE**
Aircraft	**EVACUATE**

Cabin / Electrical Fire or Smoke

LAND AS SOON AS POSSIBLE

Bat 1 / Bat 2 / Alt 1 / Alt 2 masters (memory)	**OFF**
Cabin heat / vents (memory)	**OFF / CLOSED**
Fire extinguisher	**ACTIVATE (then ventilate cabin)**

▶ Fire out and electrical power required?

Yes:

Circuit breakers	**CHECK – do not reset faulted**
Bat 2 master	**ON (essential bus – PFD, standby power)**
Bat 1 / Alt 1 / Alt 2 masters	**ON one at a time, check for smoke**
Avionics power	**ON if required**
Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable.	**NOTED**

No:

Flaps are electric: without main bus power plan a flaps-up landing (approach 90–95 KIAS, longer runway). Standby instruments and glide-range planning only.

Deferred: Electrical power off – flaps-up landing, no landing light, standby instruments.	**NOTED**

Emergency Descent

Power lever (memory)	**IDLE**
Mixture (memory)	**AS REQUIRED**
Flaps (memory)	**UP**
Airspeed (memory)	**178 KIAS G3 / 177 KIAS G5 (smooth air); 133 / 140 KIAS (turbulence)**